

3 The photograph shows jujube fruits.



(Source: © Aneeta Benwan-edam/Alamy Stock Photo)

Jujube fruits are edible and are native to China.

Jujube fruits do not store well.

An investigation compared the effect of covering the fruit with an edible coating.

One coating was made from chitosan, a polysaccharide obtained from the outer skeleton of shellfish.

Another coating was made from cinnamon oil.

Samples from stored fruit were tested for vitamin C content and antimicrobial activity.

(a) The table shows some of the results of this investigation.

Storage time / days	Vitamin C content in the fruits / a.u.		
	With no coating	Coated with chitosan	Coated with cinnamon oil
0	0.48	0.49	0.48
15	0.44	0.47	0.47
30	0.37	0.39	0.43
45	0.31	0.36	0.39
60	0.27	0.32	0.34



- (i) The percentage decrease in the vitamin C content of the fruit with no coating, over the 60-day period, was 43.75%.

Calculate the difference in the percentage decreases for the fruit coated with chitosan and the fruit with no coating.

Give your answer to two significant figures.

(2)

Answer

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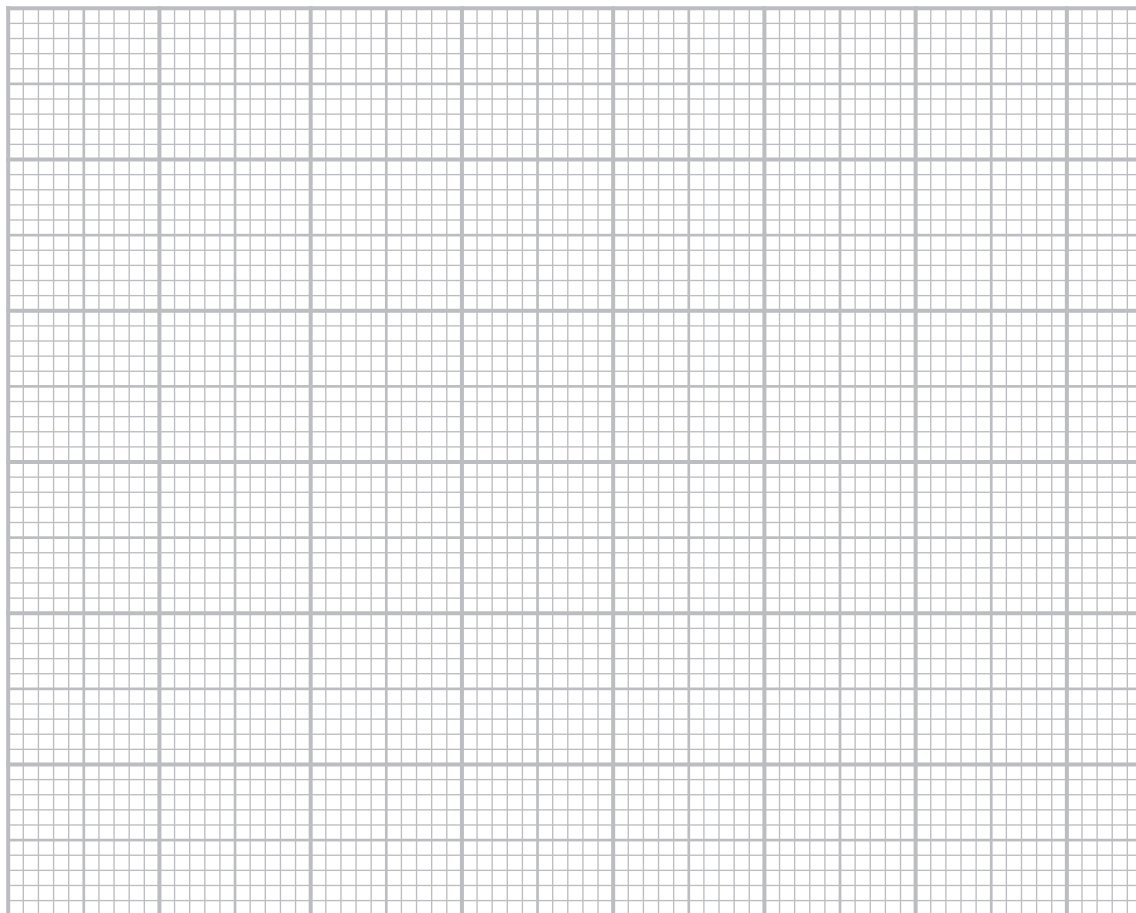
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- (ii) Plot a suitable graph to show the relationship between storage time and the vitamin C content in the fruits coated with cinnamon oil.

Draw a line of best fit through the data.

(4)



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- (iii) This data has no indication of repeatability.

Describe how data could have been collected in this investigation and presented to show its repeatability.

(3)

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- (iv) A coating made from a combination of chitosan and cinnamon oil was also tested.

Describe an investigation to determine if this coating is more effective than the two coatings used separately in preserving the vitamin C content.

(3)



- (b) The antimicrobial activity of the coatings was investigated using the agar diffusion method.

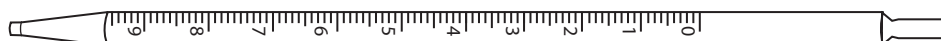
Agar plates were spread with a suspension of bacteria containing 5×10^5 bacteria per cm^3 .

Two suspensions of bacteria were made, one with *E. coli* and one with *S. aureus*.

Discs of filter paper were soaked in either the chitosan or the cinnamon oil and placed on top of the agar. The agar plates were then incubated.

Agar plates were also prepared that had two types of fungi grown on them: *A. flavus* and *P. citrinum*.

- (i) The diagram shows a graduated pipette.



The pipette was used to prepare the suspension of bacteria.

Describe how to use a graduated pipette to measure an accurate volume.

(2)

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- (ii) The suspension of bacteria containing 5×10^5 bacteria per cm^3 was used.

Each agar plate was spread with 0.2 cm^3 of this suspension.

Calculate the number of bacteria that were added to each agar plate.

Give your answer in standard form.

(1)

Answer

(iii) Explain **two** safety precautions that must be taken when preparing these plates.

(2)

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- (iv) The table shows some differences between the method using bacteria and the method using fungi.

Type of microorganism	Type of agar	Incubation temperature / °C	Length of incubation / hours
bacteria	nutrient agar	35	24
fungi	potato dextrose agar	28	72

Suggest why different conditions were used when growing these two types of microorganism.

(3)

Type of agar

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Incubation temperature

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Length of incubation

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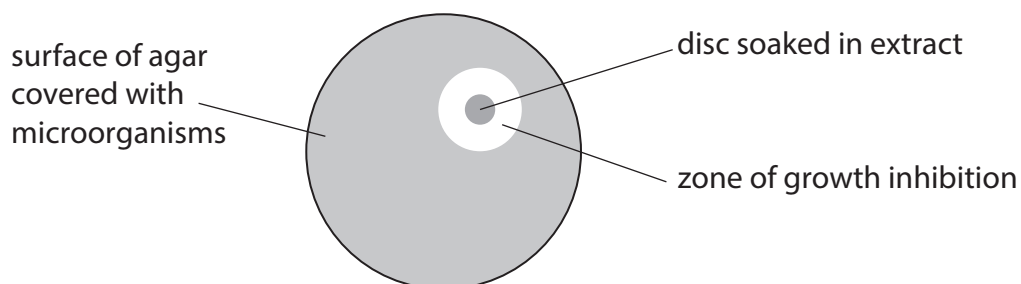
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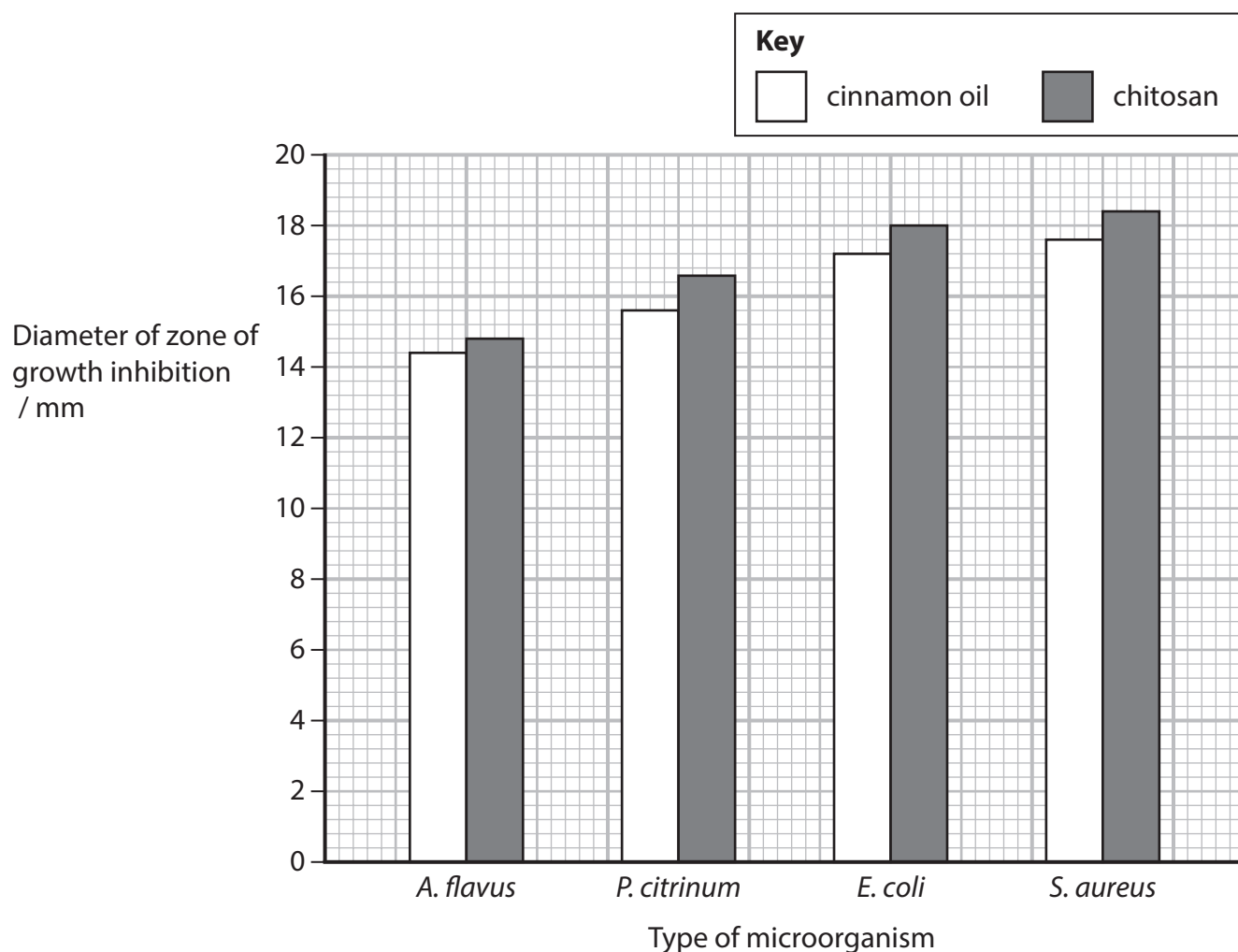


- (v) The antimicrobial activity was compared by measuring the diameter of the zone of growth inhibition. The greater the diameter, the greater the antimicrobial activity.

The diagram shows the zone of growth inhibition.



The graph shows the results of this investigation.



Evaluate the usefulness of this data when making conclusions about the antimicrobial activity of cinnamon oil and chitosan in this investigation.

(4)

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(Total for Question 3 = 24 marks)

TOTAL FOR PAPER = 50 MARKS

